

1) 10 Model Providers and Their Models

Model Provider	Model	Release Date	Cut off Date	Cost of tokens (per 1M)	Context Window	Type	When to use this model
OpenAI	GPT-4o	May 2024	Oct 2023	Input: \$5.00 Output: \$15.00	128,000 tokens	Text, Vision, Audio	Use for general everyday tasks, fast multi-modal conversations, complex logic, and standard coding help.
Anthropic	Claude 3.5 Sonnet	June 2024	April 2024	Input: \$3.00 Output: \$15.00	200,000 tokens	Text, Vision	Use when you need extreme coding precision, complex writing, logic problems, or deeply analytical research.
Google	Gemini 1.5 Pro	May 2024	Jan 2024	Input: \$3.50 Output: \$10.50	2,000,000 tokens	Text, Vision, Audio, Video	Use when you need to upload massive files like hours of video, entire code folders, or massive books all at once.
Meta	Llama 3.1 (70B)	July 2024	Dec 2023	Free (Open Source)	128,000 tokens	Text Only	Use when you want to run a powerful model on your own local servers for absolute data privacy and zero token fees.
Mistral AI	Mistral Large 2	July 2024	Nov 2023	Input: \$2.00 Output: \$6.00	128,000 tokens	Text, Code	Use for multilingual European languages, complex coding tasks, and low-cost alternative enterprise apps.

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Cohere	Command R+	April 2024	Jan 2024	Input: \$3.00 Output: \$15.00	128,000 tokens	Text Only	Use specifically for corporate business environments connecting to internal databases (RAG setup) across 10+ languages.
DeepSeek	DeepSeek-V2.5	Sept 2024	June 2024	Input: \$0.14 Output: \$0.28	128,000 tokens	Text, Code	Use when you need extremely high-quality coding or math calculations at the lowest price possible in the market.
Microsoft	Phi-3.5-MoE	Aug 2024	May 2024	Free (Open Source)	128,000 tokens	Text Only	Use when you need a small, lightweight model to run fast on local devices like a laptop or smartphone without using heavy cloud power.
Alibaba Qwen	Qwen-2-72B	June 2024	Feb 2024	Free (Open Source)	128,000 tokens	Text, Code	Use if your work involves heavy translation or logic processing across Asian languages alongside English.
xAI	Grok-2	Aug 2024	Dec 2023	Premium X Subscription	128,000 tokens	Text, Vision	Use when you need real-time up-to-date news or live information directly pulled from social media data.

2) How a Transformer Works

(a) Vector Embedding (Turning Words into Numbers)

Computers do not understand text or letters; they only understand numbers. Vector embedding is the process of translating human words into a list of numbers so the AI can process them.

Think of it like an enormous, invisible map where every single word has its own specific GPS coordinates based on its meaning. Words that are similar or related live in the same neighborhood. For example, the words "apple", "orange", and "banana" will have coordinates very close to each other on the map because they are all fruits. A completely unrelated word like "airplane" will have coordinates far away. This map allows the AI to immediately see how words relate to each other mathematically.

(b) Self-Attention Mechanism (Understanding the Context)

When you read a sentence, your brain naturally connects words together to understand what is happening. Older AI models used to read words one by one, often forgetting the beginning of a long sentence by the time they reached the end. The Self-Attention mechanism fixes this by allowing the AI to look at every single word in a sentence at the exact same time and calculate how much they depend on each other.

For example, look at these two sentences:

- "The bank of the **river** was muddy."
- "I need to go to the **bank** to deposit cash."

In the first sentence, the Self-Attention mechanism allows the word "bank" to focus heavily on the word "river," telling the AI that this is a piece of land. In the second sentence, "bank" focuses heavily on "deposit" and "cash," letting the AI know it means a financial institution. It gives words their true meaning based on their neighbors.

(c) Feed-Forward Neural Network (The Thinking and Memory Step)

Once the Self-Attention step finishes connecting the words and figuring out the context of the sentence, it passes that information to the Feed-Forward Neural Network.

Think of this as the "brainpower" or "learned knowledge" step. While Self-Attention acts like a communication line swapping clues between words, the Feed-Forward network acts like an individual processor for each word. It takes the clue-heavy words and runs them through the AI's millions of learned rules and factual memory. This is the exact step where the AI processes all those combined clues to figure out what the next best word or final answer should actually be.